

Project Proposals on Advanced Data Structures

- 1) [Study] Fibonacci heaps enable even faster algorithms for minimum spanning trees: in their original paper [8], Fredman and Tarjan give an algorithm with running time $O(m \log^* n)$, where $\log^*$ is the *iterated logarithm* function (i.e. the number of times we must take logarithms before the argument becomes smaller than 1). Study and present that algorithm.
- 2) [Study/Implementation] Kaplan et al. [11] revisited Fibonacci heaps proposing a one-tree version of this classic data structure in which each decrease-key requires only a single cut. Study and present the paper and/or implement this revisited version and compare your implementation with the original Fibonacci heaps.
- 3) [Open Problem] [11] leaves some open problems related to the analysis of their revisited Fibonacci heap. Try to solve one of those open problems. Then switch to proposal 2) if the problem evades solution.
- 4) [Implementation] A *strict Fibonacci heap* is a data structure with the same time bounds of Fibonacci heaps but in the worst case [3]. The authors also made available a Python implementation of their data structure: can you give a faster implementation?
- 5) [Study/Implementation] So you think that Dijkstra's shortest-paths algorithm implemented with a Fibonacci heap is optimal for solving the single-source shortest path problem with non-negative edge weights? Wrong. In a 2025 breakthrough, Duan et al. [5] designed a deterministic algorithm with complexity $O(m \log^{2/3} n)$, which breaks the $O(m + n \log n)$ time bound of Dijkstra's algorithm on sparse graphs, showing that Dijkstra's algorithm is not optimal for SSSP.¹ Study and present that paper. Several implementations of this algorithm are popping up: can you give a faster implementation?
- 6) [Study/Implementation] In 2024, a team of researchers [9] developed an ingenious new kind of heap that makes Dijkstra's shortest-paths algorithm *universally optimal* when we require the algorithm to produce a sorted order of the vertices by distance from the source, namely running as fast as possible on every single graph layout.² Study and present that paper (and possibly do an experimental evaluation).
- 7) [Study] A *hash table* is another ubiquitous data structure, crucial in many applications. Very recent work has introduced a new construction that achieves far better expected search complexities than were previously thought possible [6]. The main author is an undergraduate student!³ Study and present that paper.

¹Media coverage: <https://www.quantamagazine.org/new-method-is-the-fastest-way-to-find-the-best-routes-20250806/>

²Media coverage: <https://www.quantamagazine.org/computer-scientists-establish-the-best-way-to-traverse-a-graph-20241025/>

³Media coverage: <https://www.quantamagazine.org/undergraduate-upends-a-40-year-old-data-science-conjecture-20250210/>

- 8) [Study] As you saw from previous items, in the past couple of years several new algorithms for shortest paths have been discovered. The first of such breakthroughs was a near-linear algorithm for single-source shortest paths, working even in the presence of negative weights [1, 2]. Study and present that algorithm. Useful material (such as videos, lecture notes, media coverage, etc) can be found at <https://sites.google.com/site/dannanongkai/publications>
- 9) [Implementation] Very recently, Cassis et al. [4] implemented the algorithm of [2]: can you give a faster implementation?
- 10) [Study/Implementation] The shortest-path algorithms in 8) require weights to be integers; in the general case of real weights, the fastest algorithm remains the classic $O(mn)$ -time Bellman-Ford. That changed in 2024, when Fineman presented a randomized algorithm with complexity $\tilde{O}(mn^{8/9})$ [7]. Study and present that paper (or one of its several improvements, such as [10]), and possibly do an experimental evaluation.

References

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